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UNIVERSITI TEKNOLOGI MALAYSIA

SAIA 1013: Responsible AI and Ethics

MEETING 2
SECTION 01

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MEETING 2

Theoretical Concept of Ethics and their Application to AI

- Major ethical theories (e.g., Utilitarianism, Deontology, Virtue Ethics).
- Integration of Ethical Theories in AI
- Ethical theory in Decision Making

(Some are in form of these activities: Lecture, Discussion, Case Studies, Short Quiz, Self Study)

Teaching Plan Meeting 2



Class engagement session

- Online Lecture
  (Integrating Ethical Philosophy with AI Insights)



Independent Further Study (Forum):

- Select and read one more material about Ethic Theory: Consequentialism, Deontology and Virtue Ethics
- State the reference. Write summary on one main difference of these 3 theories in the forum with example in specific scenario.



Critical Thinking (Assignment 5%: Maximum 3 slides submission)

Work in a group of three or four. Select one movie or drama (short or long) that contains a storyline involving interaction between humans and Artificial Intelligence (AI). Briefly summarize the movie/drama (150–200 words).4. Identify and explain three ethical issues depicted in the story (e.g., privacy, bias, human oversight, accountability, manipulation) using reasoning. Apply at least one (1) ethics theory (e.g., Utilitarianism, Deontology, Virtue Ethics) to analyze each ethical issue you identified. Provide your conclusion: What lessons can we learn about responsible AI development and use from this movie/drama?

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RECAP ON WHAT IS GENERAL ETHICS

Ethics is knowing the difference
between what you have a right to
do and what is right to do.
"Potter Stewart"

Have you ever wondered, there are things in life permissibly legal, but it is not, right?

Why Study Ethics? (especially in this age of AI)

- Respecting Others in a Digitally Interconnected World
- Beyond “Common Wisdom”: Preparing for Complex Futures
- Shared Core Values in a Global Tech Landscape
- Rational Decision-Making in High-Stakes Environments

“The core of Ethics is to provide bases to make the best rational decisions"

What is AI Ethics?

AI ethics is about thoughtfully evaluating the moral, legal, and social issues involved in creating and using artificial intelligence systems. It's a field of applied ethics that:

- Explores how AI impacts our social, legal, and moral systems.
- Reflects on what principles, rules, and policies we should adopt to ensure those systems are fair, safe, and responsible.

Responsible AI VS AI Ethics?

Ethics Reasoning Basic

- Ethics Reasoning: Ethical reasoning is the process of thinking through what is right or wrong, good or bad, in each situation—and then deciding based on moral principles.
- It's how we justify our actions when faced with dilemmas that involve values, responsibilities, or consequences.
- Reasonably, people generally know the difference between helping and harming.
- Decisions differ because of personal values, culture, politics, or religion.

Responsibility & Capability in AI

Understanding the Moral Weight of Intelligent Systems:

- Accountability Matters: As AI systems make decisions, questions of responsibility and liability become critical who answers when things go wrong?

- What Ethical AI Requires:

✓ Defining ethical constructs/principles: What values should guide AI behavior?

✓ Implementing reasoning processes: How can machines evaluate moral choices?

Simple Example in Reasoning

- Imagine you're a manager deciding whether to fire an employee who's underperforming due to personal issues. Ethical reasoning would involve:

✓ Considering the fairness of the decision
✓ The impact on the employee's life
✓ Your duty to the company and other staff
✓ Whether support or accommodation is possible

- Ethical reasoning doesn't always lead to easy answers, but it helps ensure decisions are thoughtful, principled, and justifiable

Ethical Differences & Theories

- **Similar situations** → different choices due to individual interpretations
- Philosophy offers ethical theories to explain what counts as “ethical behavior”
- Initially, ethical reasoning draws from these theories to guide decision-making.

Three Major Branches of Ethics

Philosophical Perspective:

Branch	Focus Area
Meta-Ethics	Investigates the **origin, meaning, and role** of ethical principles
Applied Ethics	Examines **real-world issues** like euthanasia, animal rights, AI behavior
Normative Ethics	Explores **how things ought to be**; develops rules for moral conduct

Source: Dignum, V. (2019). *Responsible artificial intelligence: how to develop and use AI in a responsible way* (Vol. 2156). Cham: Springer.

How On AI?:

- ✓ *Meta Ethics* is fundamentally about reasoning, especially the kind that digs into the nature, origin, and justification of ethical thought itself
- ✓ *Applied Ethics now includes AI and robotics as key areas of concern*
- ✓ *Normative Ethics helps guide the design of ethical AI systems*

AI as Moral Agents

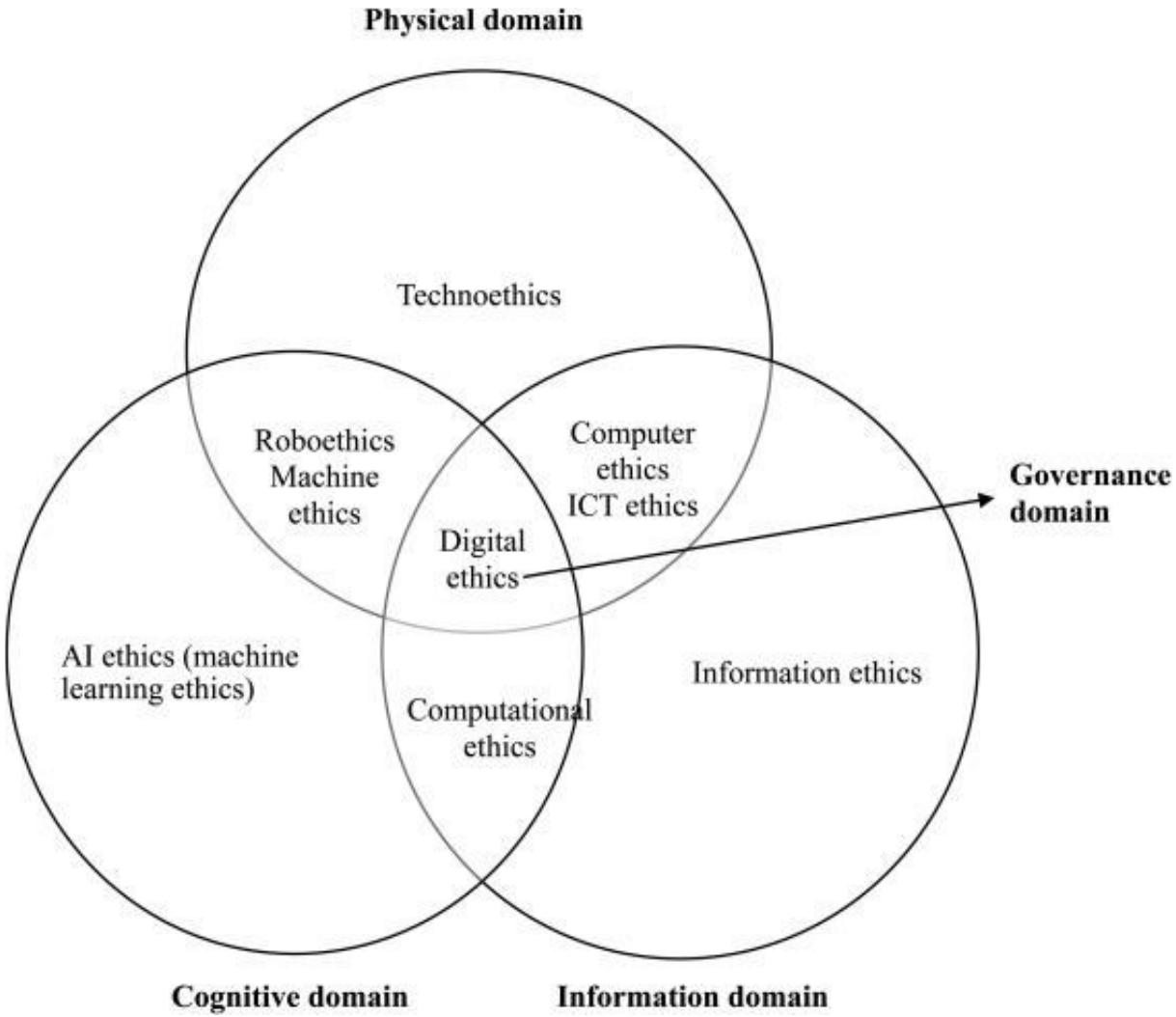
- Focus: AI making decisions with ethical impact (like humans).

- Unlike those old days, nowadays, AI is perceived as moral agents due to:

✓ Increased intelligence
✓ Autonomy
✓ Human-like interaction

- Users expect AI to respect values, human rights, and share responsibility..

Ethics in Digital Era



Physical Domain: Ethics applied to real-world, physical technologies, leading to **Technoethics**.

Cognitive Domain: Ethics addressing AI and machine learning technologies, referred to as **AI ethics** or **machine ethics**.

Information Domain: Ethics concerning information systems, covering **Information ethics** and **ICT ethics**.

Governance Domain:
Highlighting the role of governance in overseeing and regulating ethical practices across all domains.

Roboethics: Ethical considerations for robots and machines (Physical + Cognitive).

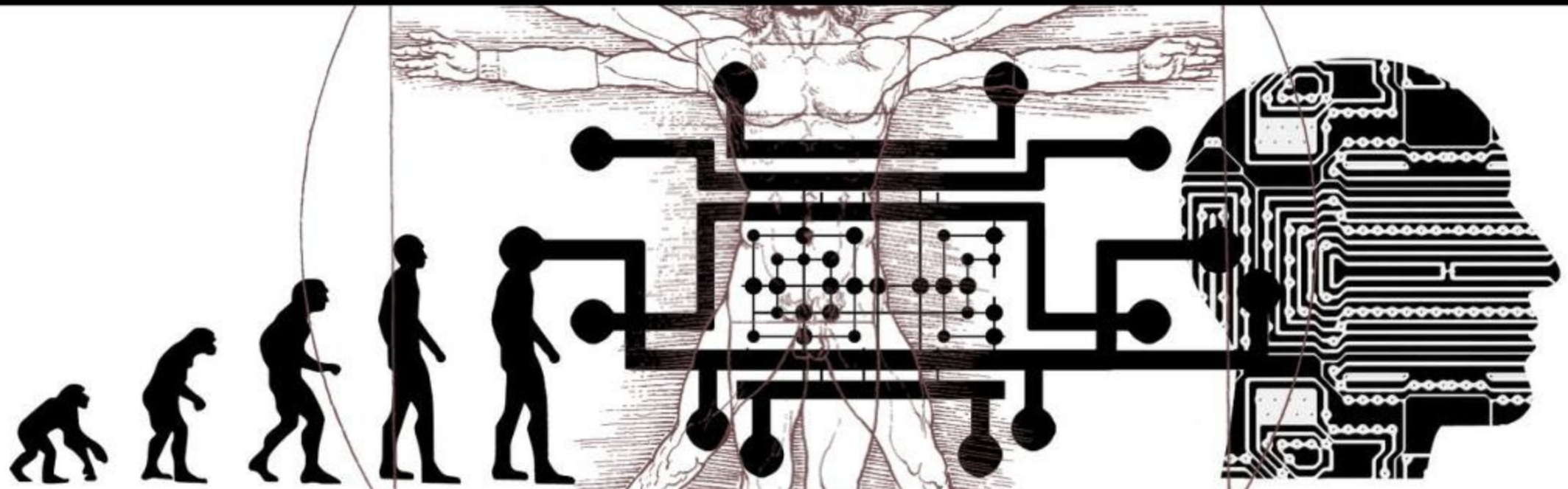
Computer Ethics/ICT Ethics: Ethical issues in computing systems and communication technologies (Physical + Information).

Computational Ethics: The ethics of algorithms and decision-making systems (Cognitive + Information).

Digital Ethics: Central to all three domains, addressing ethical dilemmas arising at the intersection of physical systems, AI, and information.

Artificial Intelligence Philosophy AI-Ethics

Theory of Ethics in AI



by Murat Durmus (@CEO_AISOMA)
https://www.linkedin.com/pulse/definitions-ai-ethics-murat-durmus/

Machine Learning Responsible AI AI-Regulation

by Murat Durmus

Top-to-Bottom Flow: From Abstract to Applied

Top Layer	Conceptual Foundations
Artificial Intelligence	The broad domain of intelligent machines
Philosophy	The source of ethical reasoning and moral inquiry
AI-Ethics	The intersection where philosophical principles meet AI challenges

Bottom Layer	Practical Implementation
Machine Learning	The technical backbone of modern AI
Responsible AI	The goal of aligning AI with ethical standards
AI-Regulation	Legal and policy frameworks to enforce ethical use

This structure implies a holistic approach to AI ethics, flow from theory to practice starting with philosophical inquiry and ending with real-world governance.

Ethics VS Moral

- Morals and ethics are distinct concepts.
- **Morals** are internal, personal guiding principles based on beliefs, habits, and convictions.
- **Ethics** are external societal norms and rules determined by groups, cultures, or environments.
- Morals and ethics can sometimes contradict each other

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Ethics VS Moral

- A lawyer might have a **moral conviction** that their client is guilty, but ethically, they are obligated to defend them.
- A whistleblower might feel **morally** compelled to share a secret, while **ethically**, they may be bound not to.
- Moral questions, such as the definition of gender, can lead to social disagreements
- These conflicts demonstrate the complex relationship between moral and ethical considerations

Philosophical Ethics Theories

Ethics Lense

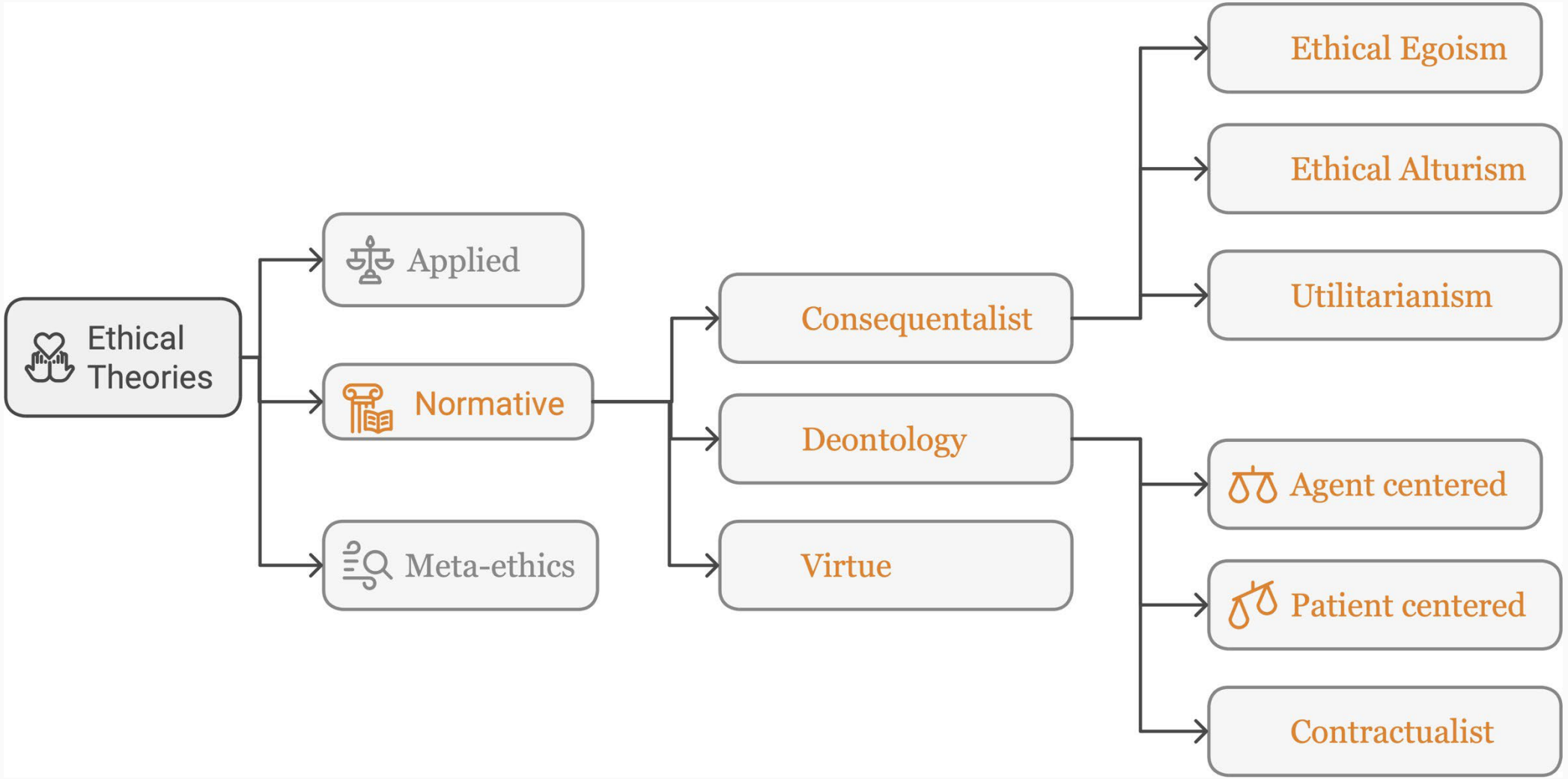
Ethics Practicality

Ethics Fundamental

Ethics Researcher

Type of Ethics	Focus	Key Questions	Example Topics
Applied Ethics	Specific moral issues like medical ethics, business ethics, AI ethics, environmental ethics.	What should we do in specific situations? Looks at how ethical principles are put into practice in real-life situations.	Medical ethics, business ethics, environmental ethics: AI: Deciding whether it is ethical to use facial recognition for public surveillance.
Normative Ethics	Moral standards and principles. Studies the rules, principles, and moral frameworks that guide what is right or wrong.	What is right or wrong? What makes actions morally good or bad?	Utilitarianism, deontology, virtue ethics AI: Using fairness principles to ensure algorithms don't discriminate.
Meta-ethics	Explores the nature, meaning, and origin of ethical concepts.	What does "right" mean? Are moral values objective or subjective?	Moral relativism, moral realism, emotivism AI: Asking whether concepts like "fairness" in algorithms are universal truths or human-made social constructs.

Branch of Ethics Theories



Modified from : An Overview of Artificial Intelligence Ethics, IEEE TRANSACTIONS ON ARTIFICIAL INTELLIGENCE, VOL. 4, NO. 4, AUGUST 2023

Key Normative Theories in AI Ethics

Theory	Core Idea	Relevance to AI
Consequentialism	Focuses on outcomes and consequences	AI should maximize beneficial results
Deontology	Based on duties and moral rules	AI must follow ethical principles
Virtue Ethics	Emphasizes character and integrity	AI should reflect virtuous behavior

Source: Dignum, V. (2019). *Responsible artificial intelligence: how to develop and use AI in a responsible way* (Vol. 2156). Cham: Springer.

Consequentialism (Teleological Ethics)

Core Key: "The Ends Justify the Means?"

- Core Principle
  - The morality of an action depends on its outcome or result
  - The morally right action is the one that produces the best overall consequences
- Key Questions in Consequentialism
  - What kind of consequences count as "good"?
  - Who is the primary beneficiary of moral action?
  - How—and by whom—are consequences judged?
- Another version of Consequentialism is Utilitarianism
  - Utilitarianism states that the best action is the one that maximizes sentient entities (well-being or happiness of affected individuals)

Deontology (Duty-Based Ethics)

Core Key: "Do What's Right—No Matter the Outcome"

- Core Principle
  - Morality is judged based on rules and duties, not consequences
  - Focuses on whether an action itself is right or wrong
  - Emphasizes moral obligation
- Key Features in Deontology are decisions should consider:
  - One's moral duties
  - The rights of others
- Often called Duty Theories
- Seen as a top-down approach: morality is guided by foundational principles
- Follow principles, even when outcomes are unfavorable

Virtue Ethics (Character-Based Ethics)

Core Key: "Be Good—Not Just Do Good"

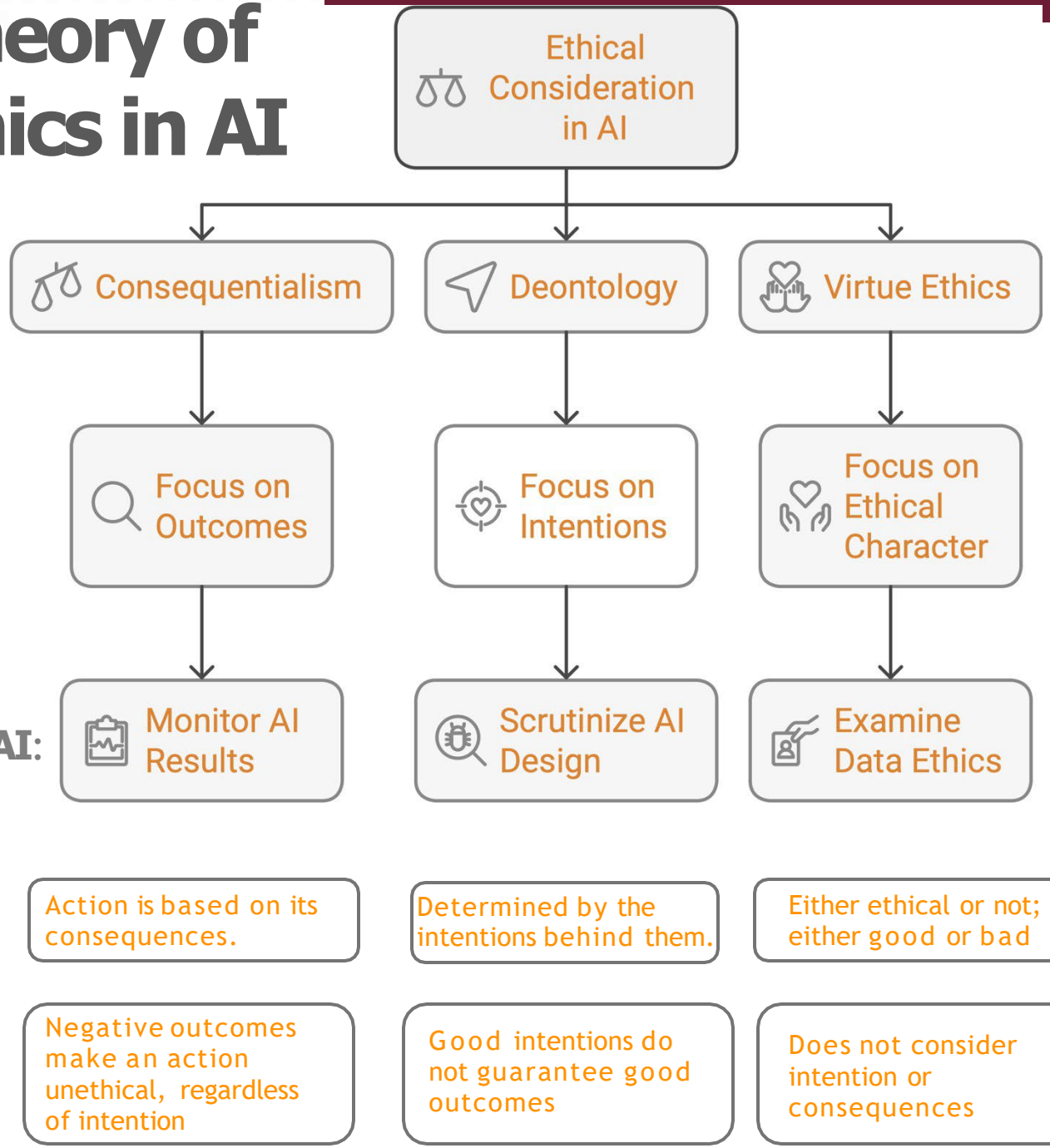
- Core Principle
  - Focuses on the character of the person, not the action or its consequences
  - Stresses the development of good habits like benevolence and courage
- Key Features in Virtue Ethics:
  - Provides practical wisdom for resolving conflicts between virtues
  - Leads to happiness and the good life.
- Aristotle's View: Human flourishing or happiness through virtuous living
- Virtue ethics evolved to include spiritual and moral dimensions

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Theory of Ethics in AI



- All three aspects are important
- It is essential to examine data for inherent biases and values.
- Intentions behind AI design and application must be thoroughly vetted.
- The outcomes and potential risks linked to the use of AI must be continually monitored

Consequentialism or Utilitarianism: Whether an action is morally right or wrong depends entirely on the consequences or outcomes of that action.

Deontologikal : Related to principles and rules that concern what ought to be done, rather than what is to be shaped

Virtue: "That duty or obligation is the basis of right and wrong."

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Comparison of the Three Normative Ethical Theories

Ethical Theory	Description	Deliberation Focus	Decision Criteria	Practical Reasoning
Virtue Ethics	An action is right if it is what a virtuous person would do in the situation.	Motives (Is action motivated by virtue?)	Virtues	Instantiation of virtues / human qualities
Deontological Ethics	An action is right if it is in accordance with a moral rule or principle.	Action (Is action compatible with some imperative?)	Duties/rules	Follow the rules
Consequentialist Ethics	An action is right if it promotes the best consequences, i.e., maximizes happiness.	Consequences (What is outcome of action?)	Comparative well-being	Maximization of utility or happiness

J. Skorin-Kapov, "Ethical positions and decision-making," in *Professional and Business Ethics Through Film*, J. Skorin-Kapov, Ed., New York, NY, USA: Springer, 2018, pp. 19–54.

Features of the Three Approaches for Implementing Ethics in AI

Approach	Description	Features			
		Require ethical rules or not ?	Learning Ability	Adaptation Ability	Interpretability
Top-Down	Program the given ethical theory and principles	Yes	No	Weak	High
Bottom-Up	Learn the general rules from individual cases	No	Strong	Strong	Low
Hybrid	Combine bottom-up and top-down approaches	Yes	Strong	Strong	Median

T.-L. Gu and L. Li, "Artificial moral agents and their design methodology: Retrospect and prospect," *Chin. J. Comput.*, vol. 44, pp. 632–651, 2021

Other Views in Normative Ethics

Beyond the Main Theories (a)

Principle of Double Effect (DDE)

- Core Idea: Deliberately causing harm is morally wrong—even if it leads to good outcomes.
- Key Distinction: Harm may be acceptable if not intended, but merely a side effect of doing good.
- Focus: DDE emphasizes intention over the act itself or its consequences.
- Moral Implication: It's harder to justify an action with a harmful effect if that effect is intended, than if it's unintended.

Other Views in Normative Ethics

Beyond the Main Theories (b)

Human Rights Ethics

- Core Principle: Human Dignity
  - Every person has immeasurable value.
  - Each life is equally valuable.
  - No life should be sacrificed for the sake of others.

- Real-World Case:
  - German Aviation Law allowed shooting down hijacked planes to save others. This meant intentionally killing innocent passengers. German court ruled it unconstitutional as it violated human dignity, and contradicted the core value of their legal system

Other Views in Normative Ethics

Beyond the Main Theories (c)

Principle of Lesser Evils

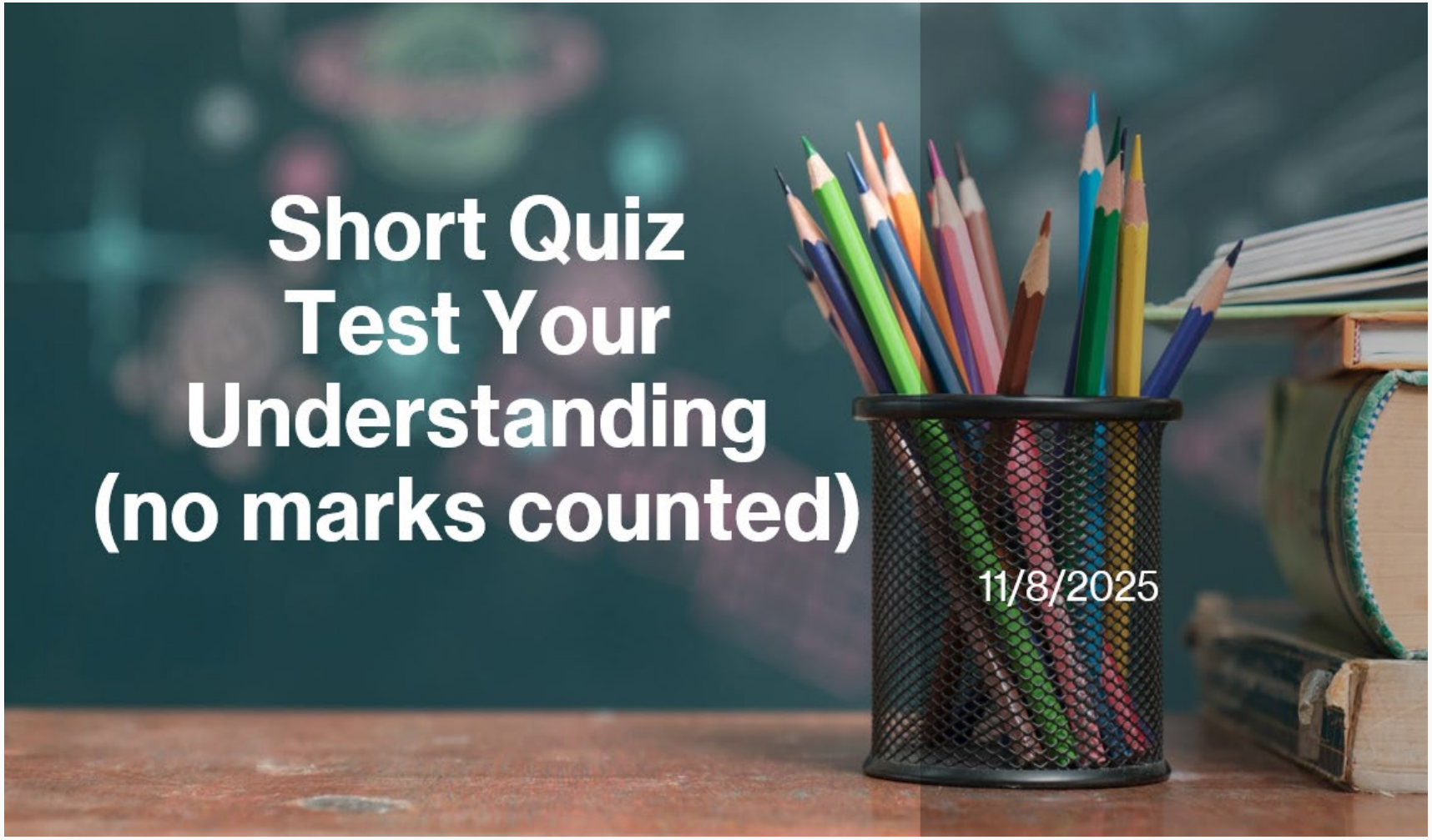
- A Consequentialist Approach
  - In moral conflicts, one may need to violate a moral rule.
  - The goal: Choose the lesser evil to minimize harm.

- Example Scenario:
  - If faced with: Lying VS. Helping a murderer. This principle suggests: “Lie”, because it causes less harm.

- However, moral values can't be neatly ordered. Ranking evils may oversimplify complex ethics

QUIZ

FIVE QUESTIONS



Key Stakeholders in AI

Stakeholder Group	Role & Responsibilities
Developers / Engineers	Design, program, and build AI systems. Follow ethical guidelines, safety standards, and transparency practices.
Researchers & Academics	Advance AI theory, algorithms, and ethical frameworks. Set standards for testing, verification, and validation.
Business Leaders & Decision Makers	Fund, own, and deploy AI solutions. Make policy, governance, and commercialization decisions.
Policymakers & Regulators	Represent governments, regulatory bodies, and standards organizations (e.g., IEEE). Create and enforce laws, policies, and ethical requirements.
End Users	Use AI products or services directly. Influence adoption and safety through trust, expectations, and feedback.
Affected Communities / Society at Large	Impacted indirectly by AI decisions (e.g., job displacement, bias, public safety). Includes vulnerable and marginalized populations.
Auditors & Oversight Bodies	Independently evaluate AI's compliance, fairness, and transparency.

Stakeholders Ethical Focus

1. Developers & Engineers
Who they are: Software engineers, data scientists, AI researchers, system architects.
Role: Design, build, and test AI algorithms, models, and systems.
Ethical focus: Transparency, bias mitigation, safety, explainability.

2. End-Users
Who they are: Individuals or organizations directly interacting with AI tools (e.g., consumers using chatbots, doctors using AI diagnostics).
Role: Operate and rely on AI for decision-making or daily tasks.
Ethical focus: Usability, accessibility, informed consent, data protection.

3. Affected Individuals & Communities
Who they are: People impacted by AI decisions, even if they do not directly use the system (e.g., applicants in AI-driven recruitment).
Role: Passive recipients or indirect stakeholders of AI outcomes.
Ethical focus: Fairness, non-discrimination, societal well-being.

Stakeholders Ethical Focus

4. Organizations & Businesses
Who they are: Companies deploying AI solutions, start-ups, corporations integrating AI into products/services.
Role: Implement and monetize AI systems.
Ethical focus: Corporate responsibility, compliance with laws and ethics, sustainable impact.

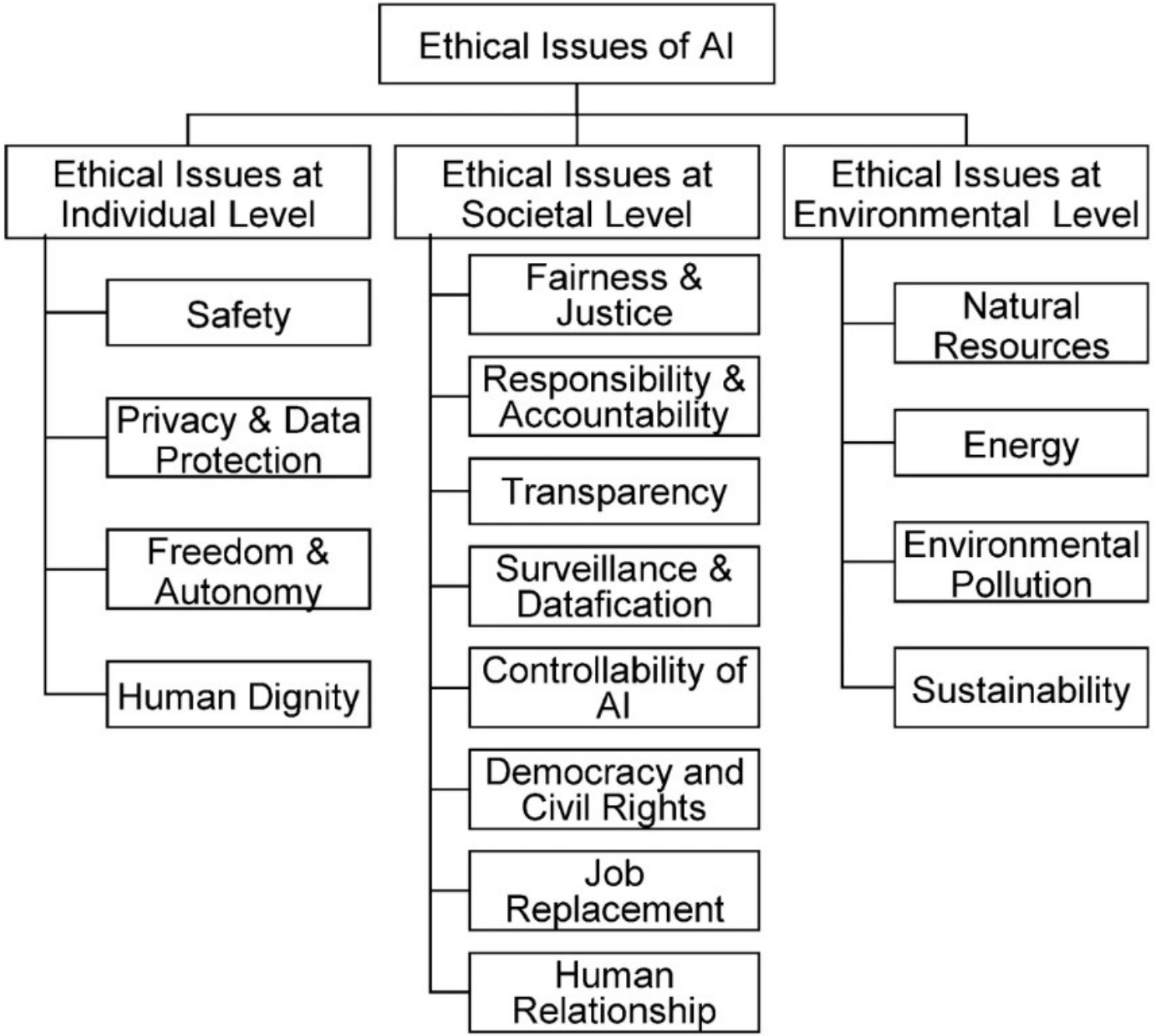
5. Regulators & Policymakers
Who they are: Government agencies, lawmakers, international bodies.
Role: Establish AI-related regulations, policies, and standards.
Ethical focus: Public safety, rights protection, accountability frameworks.

6. Academia & Researchers
Who they are: Universities, think tanks, independent research institutions.
Role: Study AI's societal, technical, and ethical implications.
Ethical focus: Knowledge dissemination, open science, unbiased research.

Stakeholders Ethical Focus

7. Civil Society & Advocacy Groups
Who they are: NGOs, advocacy organizations, consumer rights groups.
Role: Represent public interest, raise awareness of AI's impacts.
Ethical focus: Human rights protection, digital inclusion, advocacy for marginalized groups.

8. Standards Bodies & Professional Organizations
Who they are: IEEE, ISO, NIST, ACM.
Role: Create technical and ethical standards for AI.
Ethical focus: Interoperability, global ethical consensus, professional codes.


Ethical Issues of AI
Ethical Issues at Individual Level
Ethical Issues at Societal Level
Ethical Issues at Environmental Level
Safety
Privacy & Data Protection
Freedom & Autonomy
Human Dignity
Fairness & Justice
Responsibility & Accountability
Transparency
Surveillance & Datafication
Controllability of AI
Democracy and Civil Rights
Job Replacement
Human Relationship
Natural Resources
Energy
Environmental Pollution
Sustainability

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Ethical issues surrounding AI: **Individual Level**

- **Ethical Issues at Safety**
  - *Example*: Think about self-driving cars. If a self-driving car malfunctions and causes an accident, who is responsible? This raises important questions about the safety of AI technology in our daily lives.
- **Privacy & Data Protection**
  - *Example*: When you use social media, your data is collected and analyzed. Have you ever noticed targeted ads based on your recent searches? This shows how AI uses your data, and it raises concerns about how much privacy you really have.
- **Freedom & Autonomy**
  - *Example*: Smart assistants like Siri or Alexa help us with tasks, but do they also influence our choices? For instance, if a smart assistant suggests products based on your preferences, it could limit your freedom to explore other options.
- **Human Dignity**
  - *Example*: AI in hiring processes can sometimes favor certain candidates over others based on biased data. This could undermine the dignity of applicants who are equally qualified but unfairly overlooked.

Ethical issues surrounding AI: **Societal Level**

- **Fairness & Justice**
  - *Example*: Consider AI algorithms used in criminal justice to assess bail. If these algorithms are biased against certain communities, it can lead to unfair treatment and perpetuate systemic inequalities.
- **Responsibility & Accountability**
  - *Example*: If a robot makes a mistake in a hospital, such as administering the wrong medication, who is held accountable—the robot's creators, the healthcare providers, or the hospital? This raises crucial questions about responsibility.
- **Transparency**
  - *Example*: When you use a recommendation system on platforms like Netflix, have you ever wondered how it decides what you should watch next? Lack of transparency in how these algorithms work can lead to mistrust.
- **Surveillance & Datafication**
  - *Example*: Think about how facial recognition technology is used in public places. While it can enhance security, it also raises concerns about constant surveillance and loss of privacy.
- **Democracy & Civil Rights**
  - *Example*: AI used in political campaigns can manipulate public opinion through targeted ads. This raises concerns about the integrity of democratic processes and the potential for misinformation.
- **Job Replacement**
  - *Example*: With the rise of automated checkout systems in supermarkets, many cashiers are losing their jobs. This leads to discussions about the future of work and how society adapts to these changes.
- **Human Relationships**
  - *Example*: Have you seen how AI chatbots are used for customer service? While they can be efficient, they may lack the human touch, affecting how we interact with businesses and each other.

Ethical issues surrounding AI: **Environmental Level**



Natural Resources

Example:

AI is increasingly used in resource management, like predicting water usage. However, if not managed ethically, it could lead to over-extraction of resources, harming the environment.



Energy

Example:

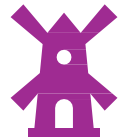
AI systems in data centers require vast amounts of energy. If these systems aren't designed efficiently, they can contribute significantly to carbon footprints.



Environmental Pollution

Example:

Consider AI in manufacturing processes. If AI optimizes for profit without regard for environmental costs, it could lead to increased pollution, affecting local ecosystems.

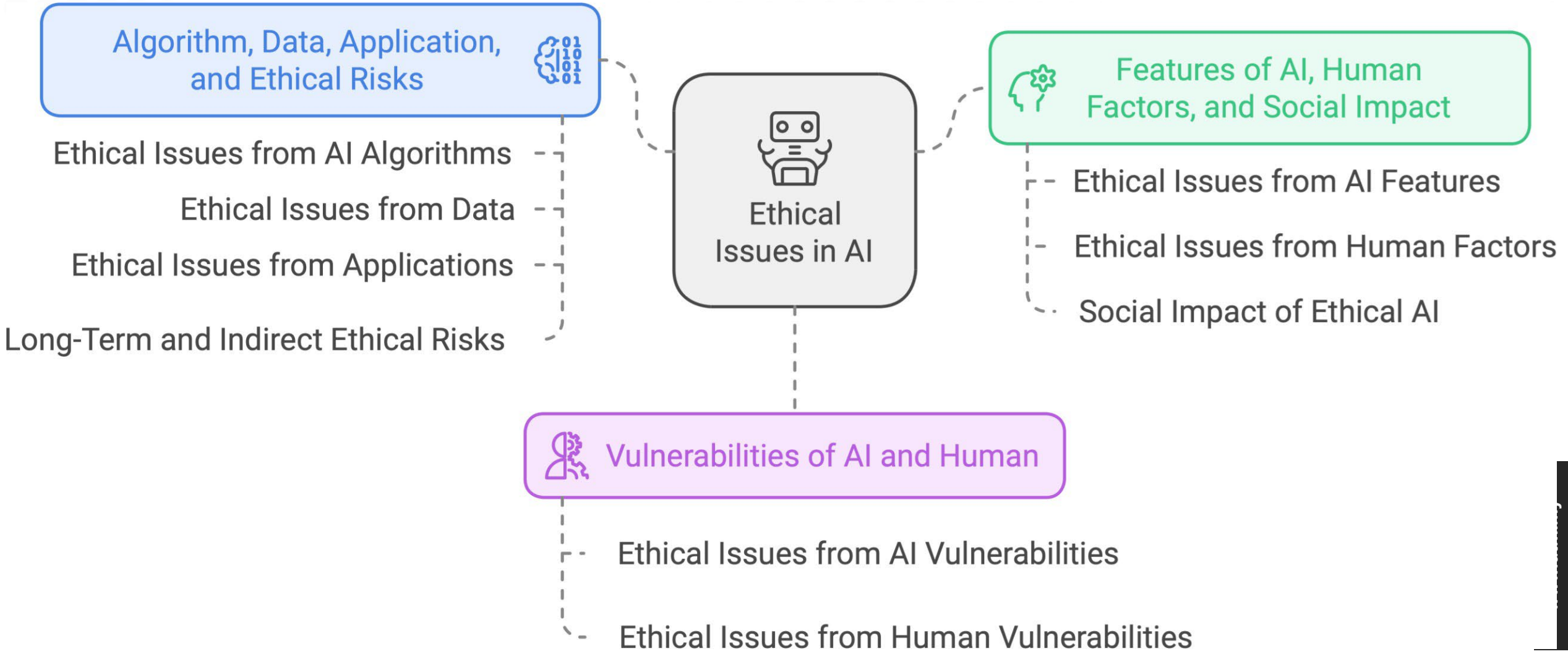


Sustainability

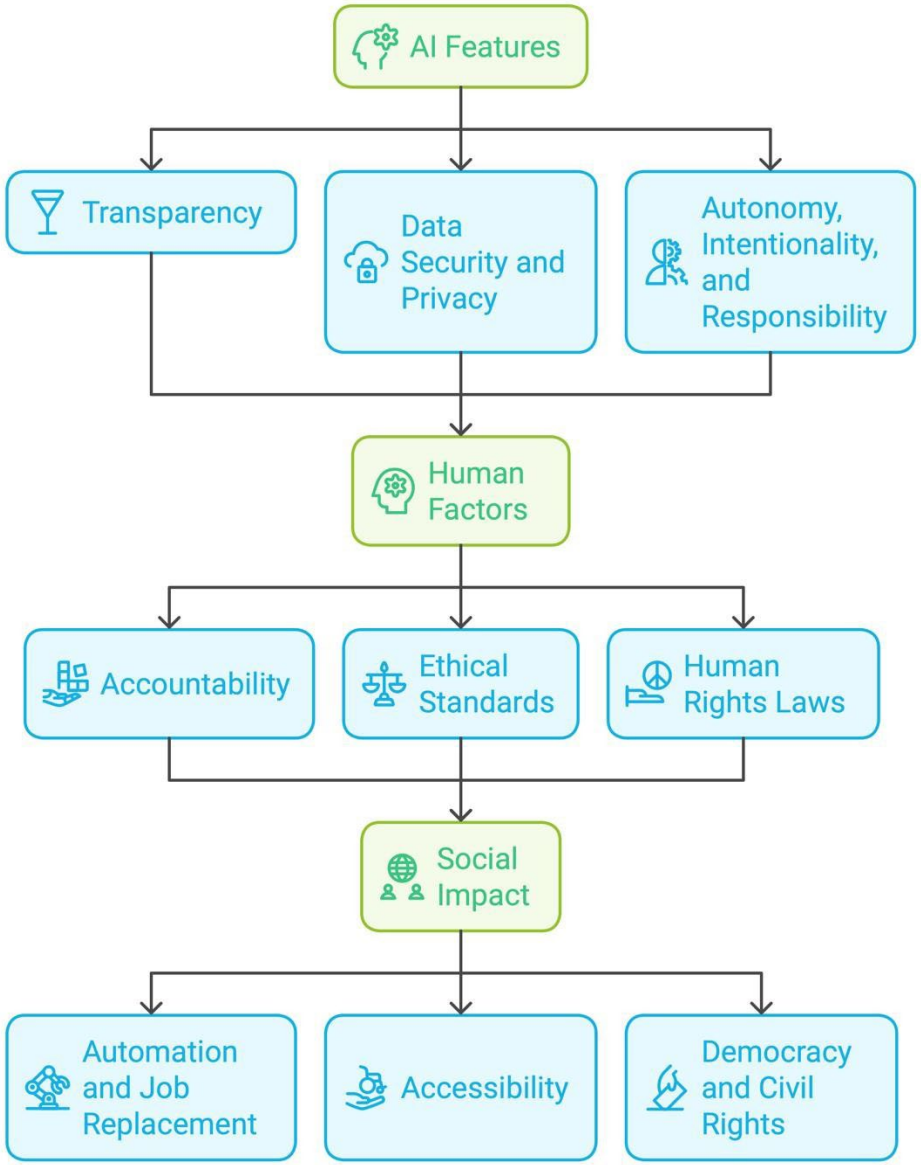
Example:

AI can help in optimizing renewable energy sources like wind and solar. However, if not used responsibly, the deployment of AI can also lead to unsustainable practices.

Categorizations of AI Ethical Issues



Categorizations Based on Features of AI, Human Factors and Social Impact

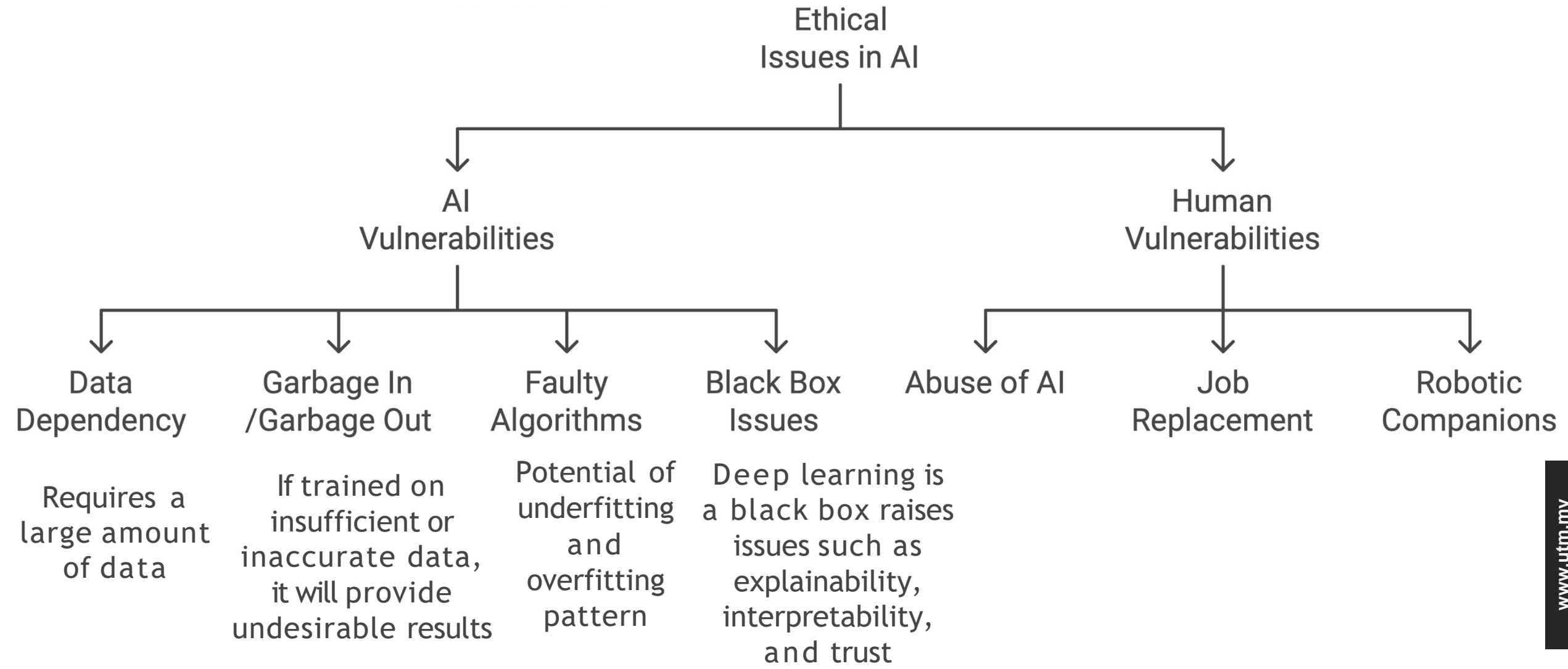


1. Ethical issues caused by features of AI
   a. Transparency: ML hard to explain and understand (black-box)
   b. Data Security and Privacy: misuse and malicious use of data
   c. Autonomy, Intentionality, and Responsibility: AI system's ability (human vs non-human, harmful vs beneficial)
2. Ethical issues caused by human factors
   a. Accountability: designing, implementation, deployment, and usage of AI
   b. Ethical Standards: comprehensive and unbiased ethical standards for training or regulating AI
   c. Human Rights Laws: training to avoid infringe and breach essential human rights
3. Social impact of ethical AI issues
   a. Automation and Job Replacement: disrupt and transform the labor market
   b. Accessibility: spread equally to everyone
   c. Democracy and Civil Rights: avoid loss of informed and trusting communities

https://www.computer.org/csdl/journal/ai/2023/04/09844014/1Fnr097UNd6

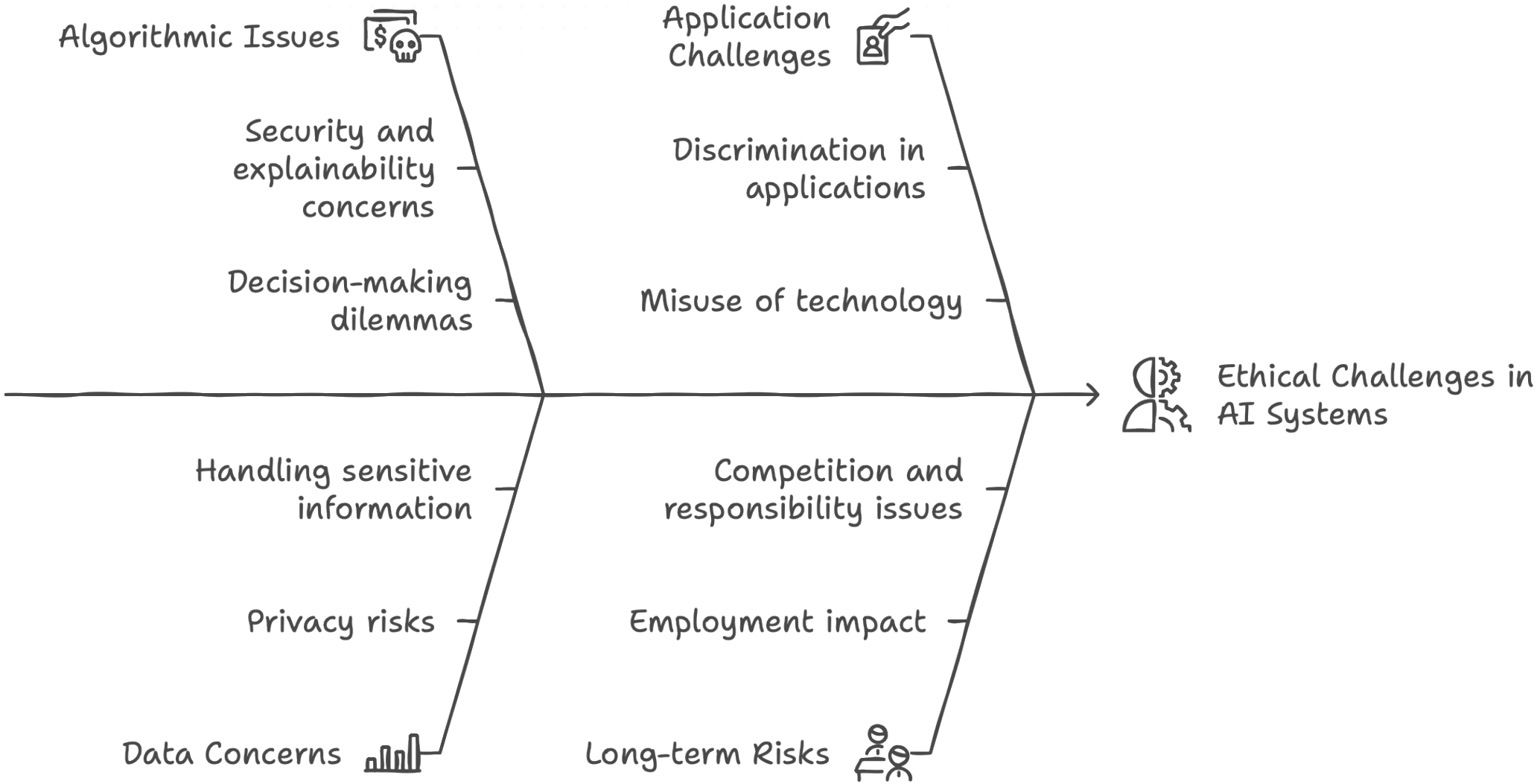
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Categorizations Based on Vulnerabilities of AI and Human



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Ethical Challenges in AI Systems

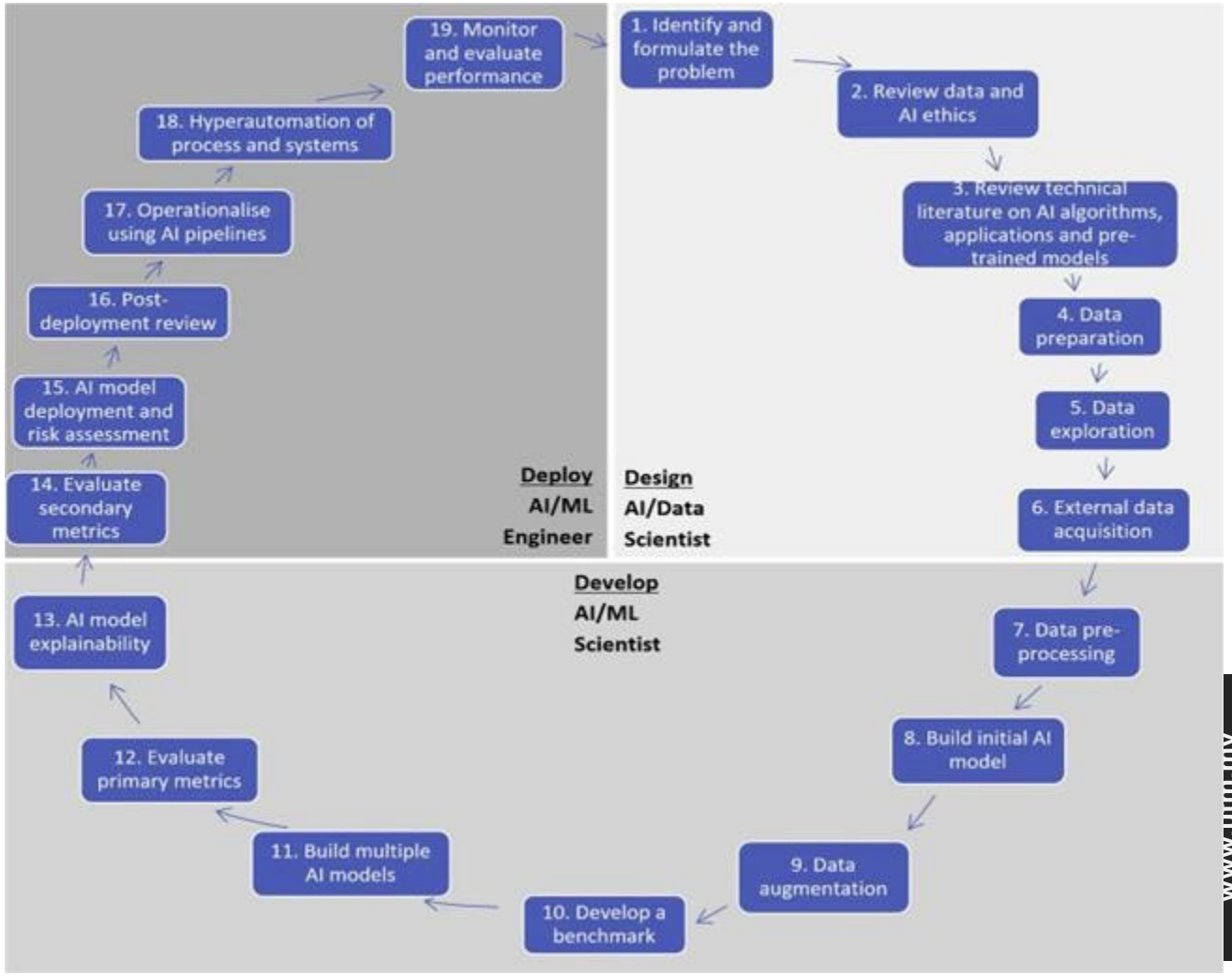


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Ethical AI Development Process

- Incorporate ethics from the design phase
- Continuous ethical assessment throughout development
- Diverse teams including ethicists, developers, and domain experts
- Regular ethical audits and adjustments

- Importance of human involvement in AI ethical decision-making
- Defining the appropriate level of human control
- Balancing efficiency of AI with ethical considerations



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Future of Ethics in AI

1. Evolving ethical frameworks as AI capabilities grow
2. Potential for AI to contribute to ethical theory development
3. Global collaboration on AI ethics standards

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Conclusion - Ethics as a Compass

- ✓ Ethics isn’t just theory; it’s a practical tool for shaping a better future.
- ✓ In a world driven by AI, ethical reasoning keeps humanity at the center.
- ✓ Studying ethics empowers us to build technologies we can trust

Assignment Critical Thinking (5%)



Independent Further Study (Forum):

- Select and read one more material about Ethic Theory: Consequentialism, Deontology and Virtue Ethics
- State the reference. Write summary on one main difference of these 3 theories in the forum with example in specific scenario.



Critical Thinking (Assignment 5%: Maximum 3 slides submission)

Work in a group of three or four. Select one movie or drama (short or long) that contains a storyline involving interaction between humans and Artificial Intelligence (AI). Briefly summarize the movie/drama (150–200 words).4. Identify and explain three ethical issues depicted in the story (e.g., privacy, bias, human oversight, accountability, manipulation). Apply at least one (1) ethics theory (e.g., Utilitarianism, Deontology, Virtue Ethics) to analyze each ethical issue you identified. Provide your conclusion: What lessons can we learn about responsible AI development and use from this movie/drama?

Assessment Criteria:

Relevance and clarity of summary/lesson learned (1%)
Identification and explanation of ethical issues with reasoning (2%)
Application of ethics theories (2%)